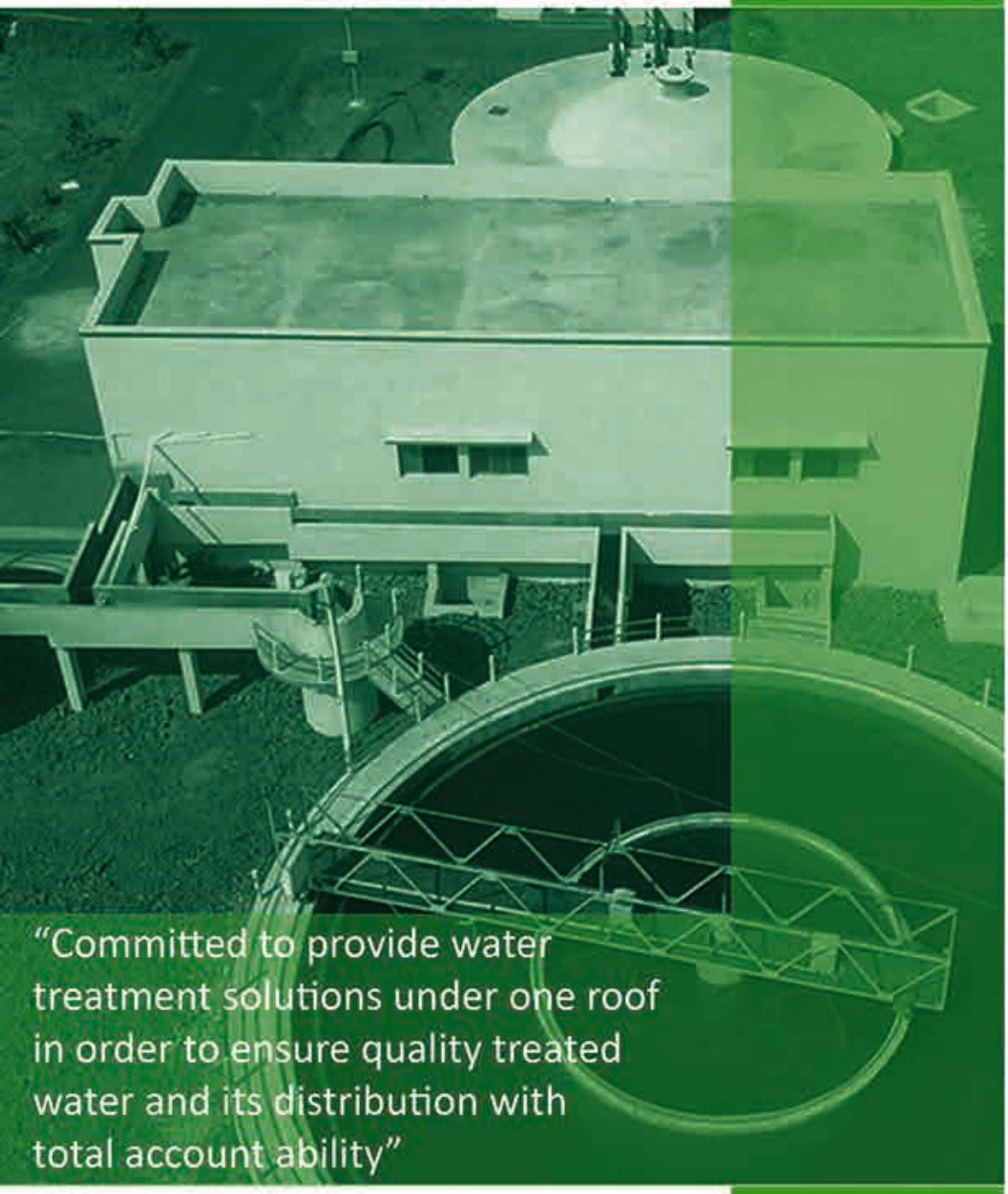
Creintors Environmental Solutions Pvt.Ltd.

Crienviro®

Engineering Eco-Sustainability



"Committed to provide water treatment solutions under one roof in order to ensure quality treated water and its distribution with total account ability"

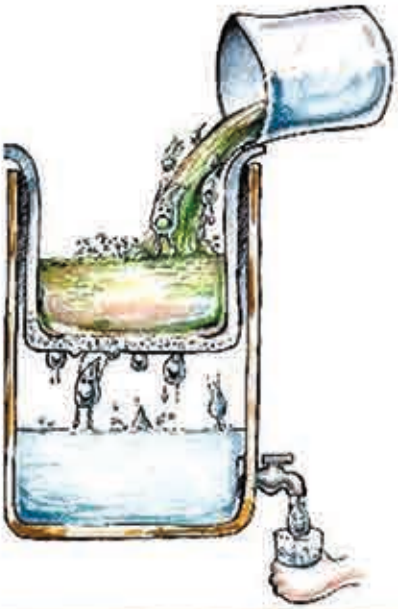
Water Treatment Solutions



www.crienviro.com



Drinking Water Treatment





Industrial Effluent Treatment



Sewage Treatment

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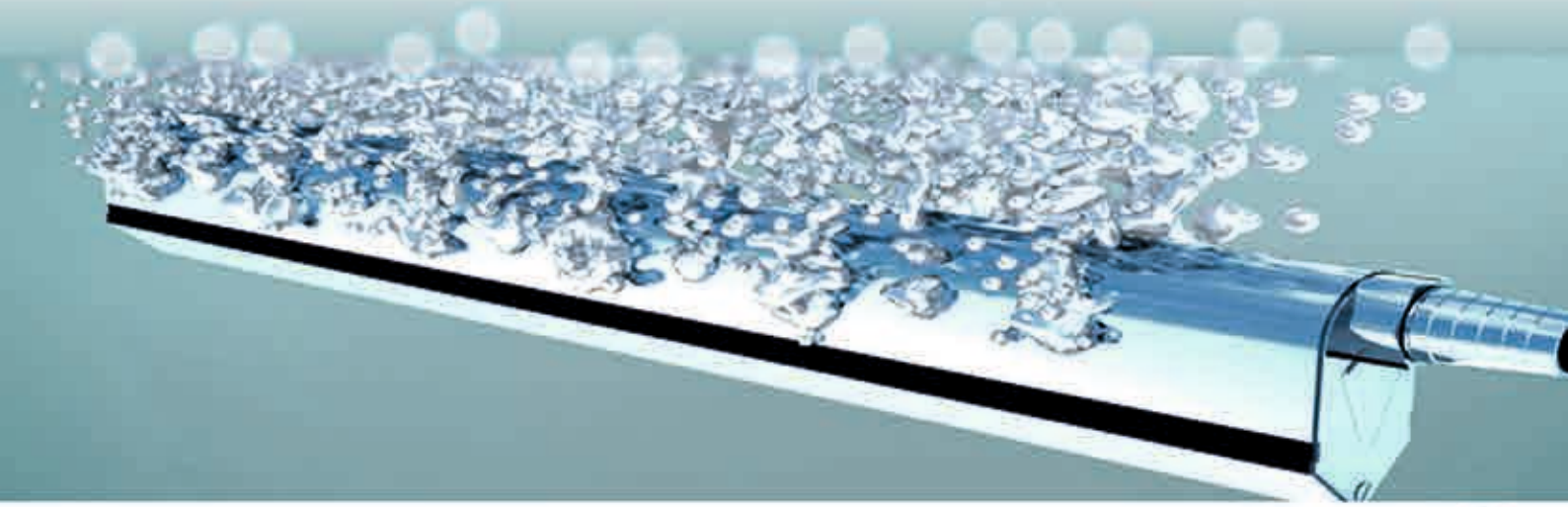
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Screening

Customized screening solutions are provided to suit the application. Based on the treatment plant the screening matter varies by size and nature. The effectiveness of the screening is important to ensure safe pumping of water as it avoids pumps, valves and other mechanical equipment from any clogging and damage. Creintors supplies different types of screens as per applications such as;

- Coarse Screens
- Fine Screens
- Grit Removal Screens

Aeration

Aeration is often the first major process at the water treatment plant. Our aeration system design ensures needed contact of water and air to remove dissolved gases and oxidizes dissolved metals such as iron, hydrogen sulfide, and volatile organic chemicals.Creintors has a range of aeration systems.

- Cascade System
- Spraying System
- Diffuser System
- Packed Column System









Mixing

Creintors mixing systems provide efficient and effective mixing mechanism to suit the system requirement such as, type of chemicals to be mixed, the inflow retention time and desired quality and quantity of discharged solution.

Our mixing solutions are provided to wide variety of applications, like the most common being the alum mixing systems and flash mixers for drinking water treatment. We have been supplying various kinds of mixers such as;

- **Pressured Water Jets**
- **In-line Static Mixer**
- **Hydraulic Mixer**
- **Mechanical Mixer**













Sedimentation

Removal of suspended particles existing in raw water via settling process with the action of coagulating agent and flocculating agent is done by means of sedimentation.

Creintors have been supplying different kinds of sedimentation equipments for water treatment mainly:

- **Clariflocculators**
- **High Rate Clarifiers**
- **Dissolved Air Flotation (DAF) Systems**

Clariflocculator

This system is designed in such a way that both flocculation and sedimentation will happen in a single unit. Creintors supply the complete electro-mechanical system ensuring efficient functioning of clariflocculator which involves rotating truss assembly, agitator assembly, slip ring assembly, control panel, etc., along with necessary control system.

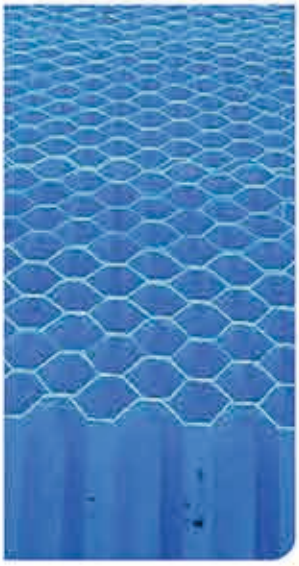


High Rate Clarifier

Creintors made High Rate Clarifier Systems provide large effective settling area for a small footprint to separate particulates from liquid. High Rate Clarifier includes Lamella Clarifier, Tube Settler and Plate Settler.







Dissolved Air Flotation (DAF)

Is a process that clarifies raw water by removal of oil, grease, algae, suspended solids and dissolved solids. The removal is achieved by micro air bubbles released from the pressured dissolved air water mixture in the flotation tank. These bubbles adhere to the suspended particles forming a floc and then floating to the surface of water in form of froth which is removed by a skimming device.

Creintors provide circular, rectangular dissolved air flotation and air dissolving tube of various sizes, from small to large water treatment plants depending on the input water characteristics.



Circular DAF

Creintors circular DAF is a specially designed COMPACT floatation UNIT which clarifies water in small retention time of 3 minutes. It consists of three concentric circular sections, the inner section is for froth collection, middle circular region for clean water collection and outer floatation vessel. A well designed scoop with a special driven mechanism is provided to remove the floated sludge and clean water is collected by extraction pipe.



Main Applications:

- Fibre Recovery
- Dairy Wastewater
- Edible Oil Effluents
- Pharmaceutical Effluents
- Slaughterhouse Effluents
- PVC Production
- Palm Oil Refining
- Waste Oil Processing
- Paint Industry Effluents
- Fertilizer Industry Effluents
- Textile Effluents, etc.

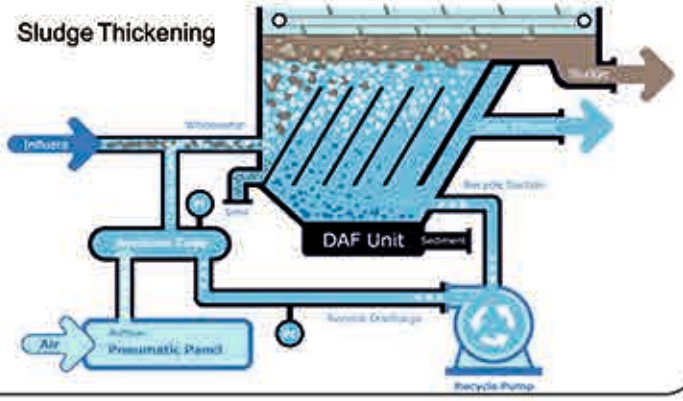


Rectangular DAF

Creintors rectangular DAF with retention time about 20 to 30 minutes, provides more uniform distribution of water and micro bubble throughout the length. It has chemical mixing unit , flocculation unit and flotation unit. The system comprises top skimming device to remove the float completely without disturbing the water layer, and bottom auger to discharge setteled sludge.

Rectangular DAF finds its applications in:

- Food Industry Effluents with high solids concentration
- Oil and fats removal without chemical treatment
- Sludge Thickening

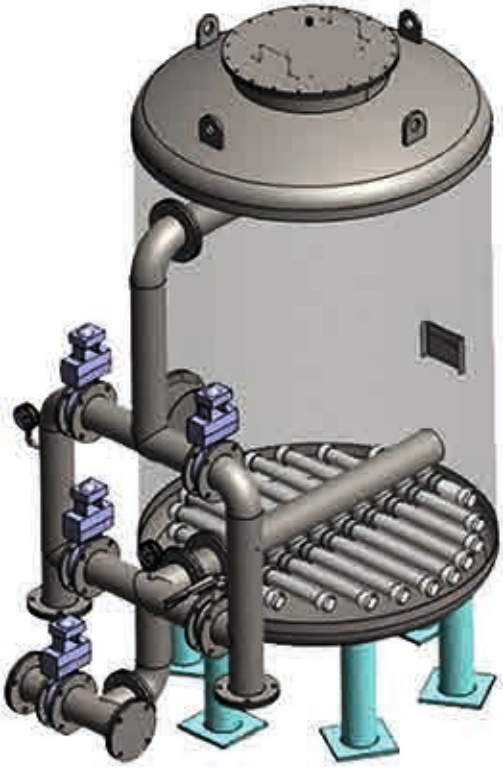






Filtration

Filtration is usually the final step in the solids removal process once the coagulation, flocculation and sedimentation process is completed. The purpose of filtration is to remove suspended particles from water by passing the water through filter media. In the filter, up to 99.5% of the suspended solids in the water can be removed, including minerals, floc, and microorganisms.

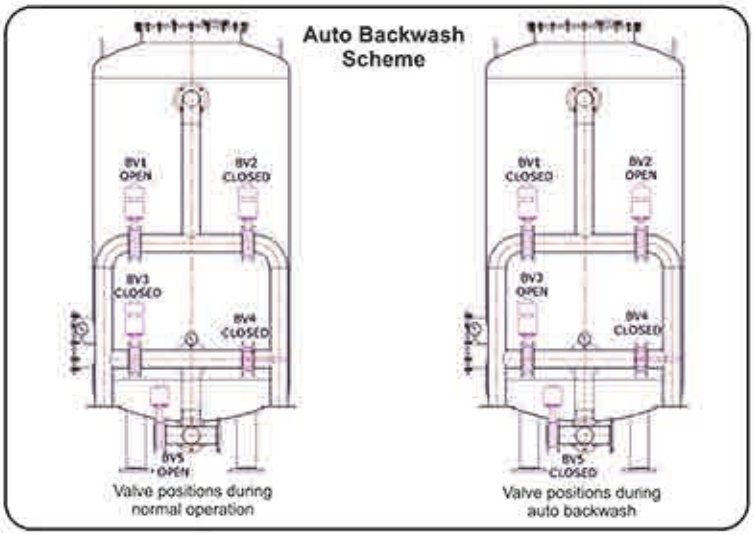


Pressure Sand Filter

Our unique design of Pressure Sand Filter with option of auto back wash ensures the maximum utilization of the surface area, lesser pressure drop across the pressure bed and effective elimination of the impurities. Creintors Pressure Sand Filters are available in different sizes to suit for smallest to large MLD plants.

The filtration media are selected and proportionally layered with graded sand, gravels, silex, silica, anthracite, and activated carbon to ensure the desired quality of water after filtration.





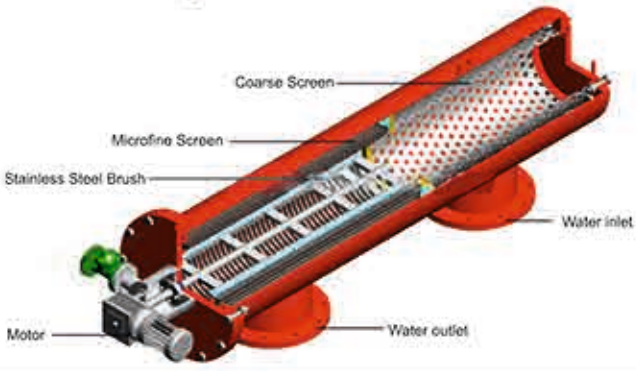
Online Hydraulic Filter

We design and manufacture customized online hydraulic filters which are automatic in operation with inbuilt self cleaning arrangement. This makes these filters provide high rate filtration and steady supply of quality water.

Fluid Discharged from Hydraulic cleaner enters through the filtration surface of filter element. The contaminants retained on the filtration surface of filter media, are flushed out through automatic backwash process.

A Controller keeps monitoring the differential pressure across the filter. The backwash cycle starts when the differential pressure reaches at a pre-set level. After backwashing for a period of time, the controller reset the filtration process. High performance and least maintenance are the key features of this unit.





Rapid Sand Filter Module

Creintors have been supplying fully automated containerized based rapid sand filter module for water purification, commonly used in drinking water facilities as a part of multiple-stage treatment system.





Membrane Filtration -RO

Creintors supply different kinds Membrane filtration unit such as Reverse Osmosis unit, Nano Filtration unit utilized for various treatment processes which includes drinking water treatment and effluent water treatment. The customized design ensures optimized and cost effective solutions.









Chemical Dosing System

We supply Chemical Dosing Systems for dosing the coagulants like Lime, Ferrous Sulphate, PAC and Alum. Creintors made Dosing systems are well designed compact systems mounted on skids and are engineered to provide steady set dosing rates for varying conditions. The dosing rate can be changed to a range to meet the dosing PPM in the flow.

Complete Dosing systems consists of Creintors Dosing Pump with corrosion resistance material of construction, mixing tanks in FRP / HDPE / SS. Creintors supplied Chemical Dosing Systems can be interface to an automation system to have online control and monitoring.



Creintors Dosing Pump

Creintors Dosing pump has been designed to deliver the exact dosage of chemicals such as Alum, Sodium Hypochlorite, etc. These pumps are developed in accordance with the set industry standards and have wide range of flow range of flow rates from 0-100 LPH. We have been supplying varying type of dosing pumps based on application such as;

- **Plunger Pump**
- **Mechanical Diaphragm Pump**
- **Hydraulic Diaphragm Pump**
- **Metering Pump**









Disinfection

Disinfection systems need to ensure right dosage and right concentration of disinfectants discharge in drinking water supply, so as to ensure safe drinking water. Both over dosage and under dosage can affect the health of users.

Based on the applications and customer requirements we offer different disinfection systems, namely;

- **Electrochlorinator Type**
- **Gas Chlorinator Type**
- **Ozonation Type**
- **UV (Ultra Violet) Type**

Electrochlorinator Type

Our well designed electrochlorinator produces hypochlorite by running an electric current through salt water. Generated hypochlorite is used to disinfect water and make it safe for human use such as for drinking water. Electrochlorinators can be provided with Batch Type or Continuous Type.



Ozonation Type

We design, manufacture and supply ozone generators incorporating the most sophisticated electronics technology available, offering wide-range of ozone generators from the laboratory scale to industrial scale. Ozonation is used for oxidation of contaminants viruses, bacteria, microorganisms and achieves excellent removal of odor and taste

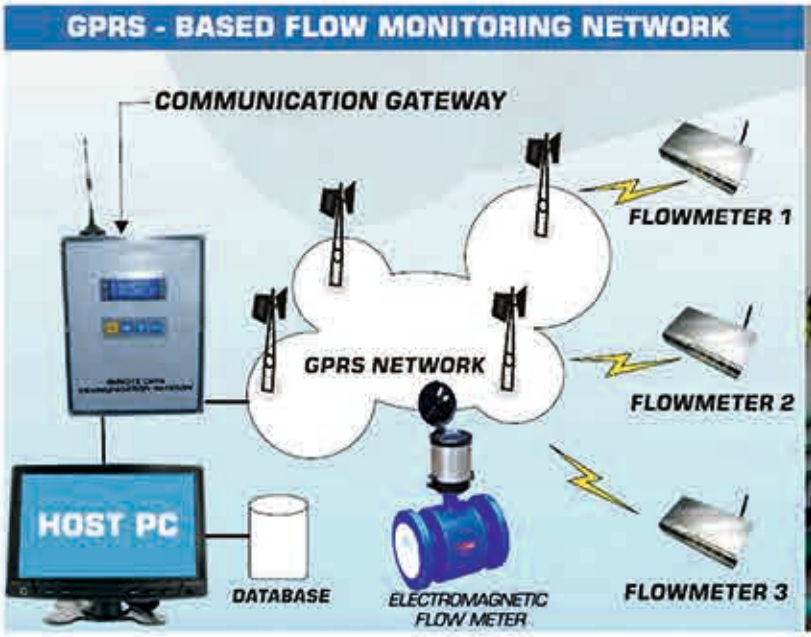
UV (Ultra Violet) Type

We provide a wide range of ultraviolet disinfection system with High-performance, energy-efficient, long-lasting UV lamps which inactivates more than 99.99% of all waterborne pathogens in just seconds.

Gas Chlorinator Type

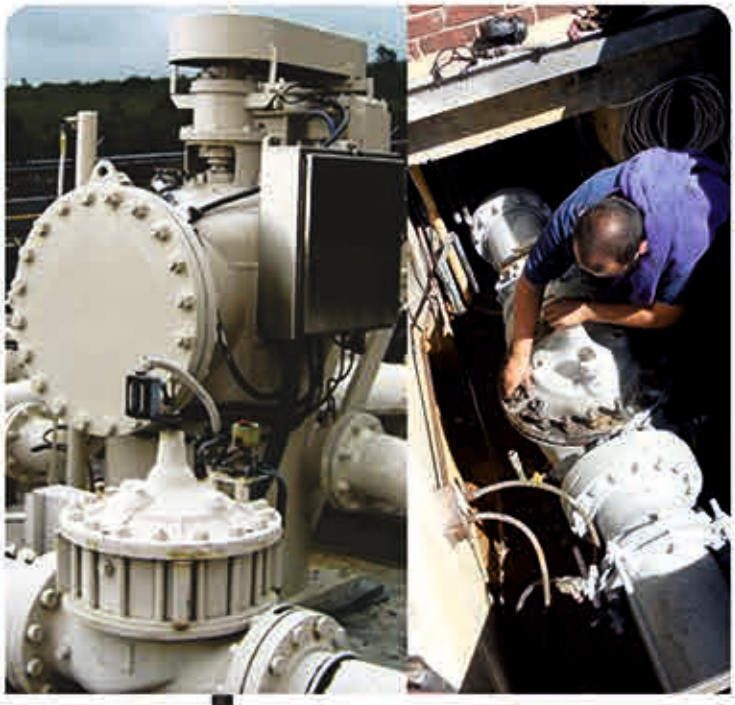
For higher range of MLD's where Electrochlorinators have high initial costs, Gas Chlorinators are used. Our unit has inbuilt all safety features to avoid any health hazards.







Water Distribution Scheme



Water distribution scheme involve equipments and automation methodology, from ensuring water in storage tanks, to pumping quantified water supply to required destinations. Our expertise in areas of sensing, pumping, flow control, measurement integrated with state of art automation have yield results to ensure customer delight.

Our pumping automation system methodology provides optimum, reliable and energy saving solutions, eliminating over huge over head storage tanks and other related civil constructions which add to huge project cost, time delays and high risk man power work.

The key to ensure required flow through all distribution piping, maintaining safe pressure for piping is achieved by using our pressure and flow balancing type regulating valves. These valves have pressure compensation features to set required flow. Ease of operation and reliability are key features of our valves.









Monitoring of flow from jackwell to aerator to treated water pumping to the MBT's and further through distribution scheme to various villages or townships has been our expertise.

Creintors design, manufacture and supply mechanical, ultrasonic, electromagnetic, AMR, AMI type and also other customized flow meters.

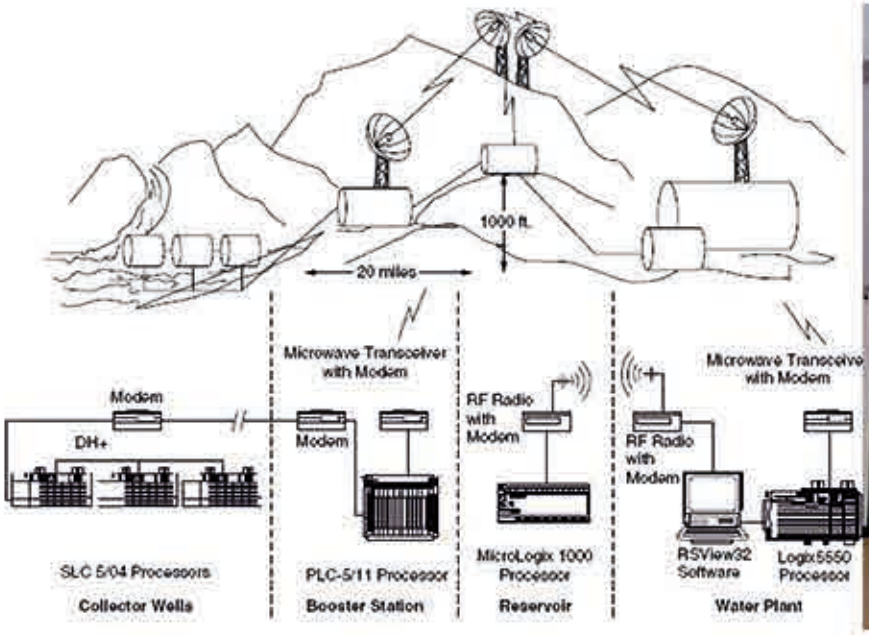
We understand the need of accountability and the reason why customers want to mount the flow meters. Only supply, installation and commissioning is not the job done, we need to provide features to have online data to mobile and mails in form of reports as desired by the customers.

Thus further data logging, sorting, presenting and storage becomes the default feature in our flow measurement systems. These are the salient features which we can provide with any of our electronic flow meter.









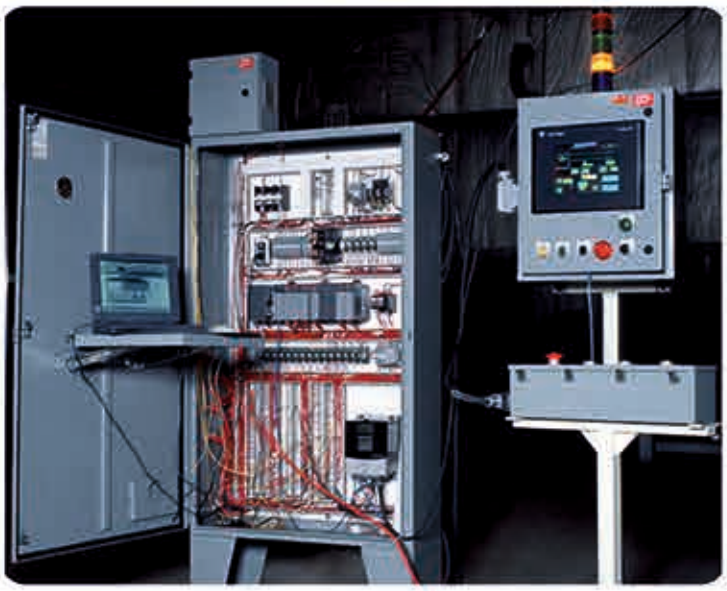


Automation

Creintors is committed to provide all water treatment solutions under one roof, with our expertise in design and manufacture of customized water treatment equipments which after integration ensure desired results. There has always been customer demand for automated and very minimal maintenance plants. This demand has always been satisfied by our expert automation and integration team.

We have access and implementation expertise on latest technologies which involves SCADA, GSM based communication and controls with state of art reporting system with data collection, storage and authorised retrieval systems. Automated mobile and computer email alerts and reports with Creintors own mobile and computer applications (apps) that work with cloud computing technologies. Where there are issues with GSM based communication network, we provide dedicated RF transmitter and receiver system that work for 2.45 Ghz. and 5 Ghz. international free band frequencies.

Our automation architecture has inbuilt redundancy and supports all mode of operations from fully automation, semi automation to fully manual systems with most simplified controls. This feature ensures no plant operation interruption and thus supply of treated water.







In today's need of fast project execution requirements, where we speak of smaller MLD's plant up to 5 MLD to be executed in few months. Then, fabrication based modular plants are the answers. They are most cost effective and more efficient by design than conventional concrete plants. Fabricated modular plants have lesser initial and maintenance cost.

Fabricated modular design has facility to add required modules to increase the capacity of treatment without affecting the plant operation. Overhead storage tank is optional as Creintors provide total pumping automation system for water distribution.



Online Plant Operation Automation System



- GSM Based
- Alerts to Mobile and PC by messages and mails
- Logging of key plant operation data
- Inbuilt battery backup of 48 hours

Online Water Quality Information System



- pH
- Turbidity
- Hardness
- Chlorides

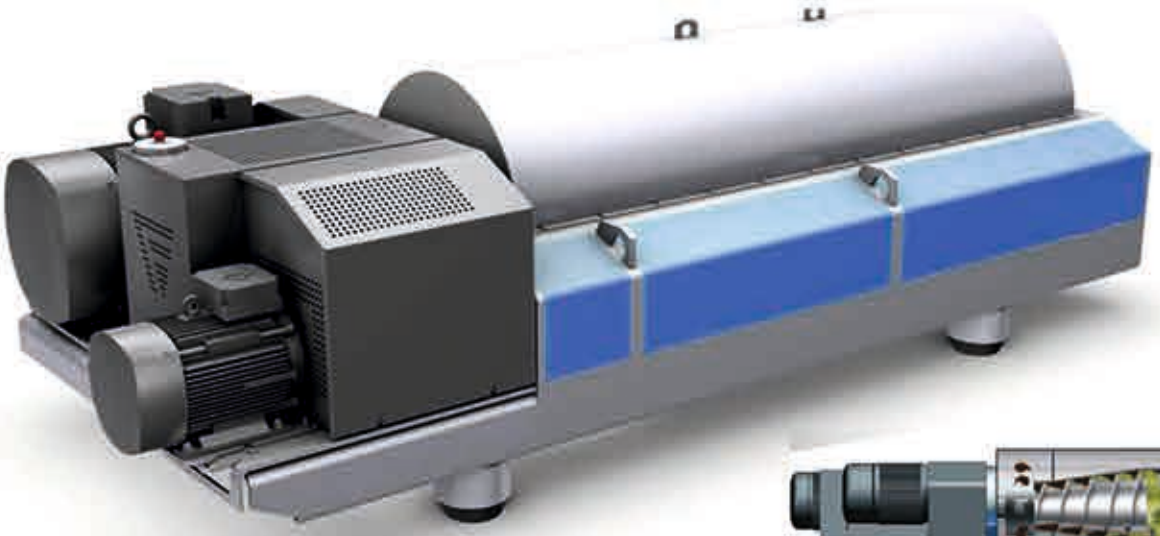
with features such as GSM based, Alerts to Mobile and PC by messages and mails. Has inbuilt battery backup of 48 hours.

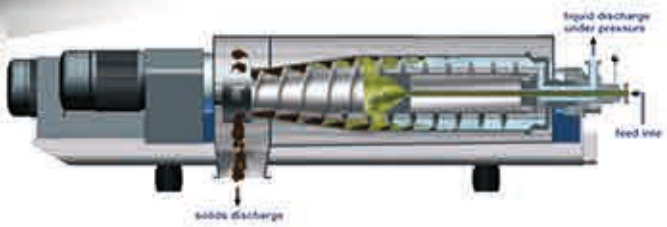
Creintors believes in prompt service and actions to keep plant working for 24/7/365 days. Our preventive maintenance checks ensure no emergencies. Our skilled and professional service team is ever ready to attend any site issues. We have our ON CALL support system which most of the time resolves certain issues and makes customer comfortable.

We always maintain inventory of spares that are required for treatment plants to ensure prompt replacement and repair. We provide training to plant operator ensuring they understand the system, operate and maintain with ease. Onsite as well as infactory training is provided.

We understand that our work in water treatment area is the best service to humanity and environment.







Solid Liquid Separation Unit

Water treatment process produces sludge that is to be dewatered prior to disposal in order to reduce disposal costs. Creintors manufactures and supplies wide range of solid liquid separation systems for dewatering of sludge such as ;

Rotary Drum Filter

A rotary drum filter consists of a large rotating drum covered by a cloth. The drum is suspended on an axial over a trough containing liquid/solids slurry. As the drum rotates into and out of the trough, the slurry is sucked on the surface of the cloth by means of vacuum and rotated out of the liquid/solids suspension as a cake.

Centrifuge

Centrifugal dewatering of sludge is a process which uses the force developed by fast rotation of a cylindrical drum or bowl to separate the sludge solids from the liquid. Density differences cause the sludge solids and the liquid to separate into two distinct layers.

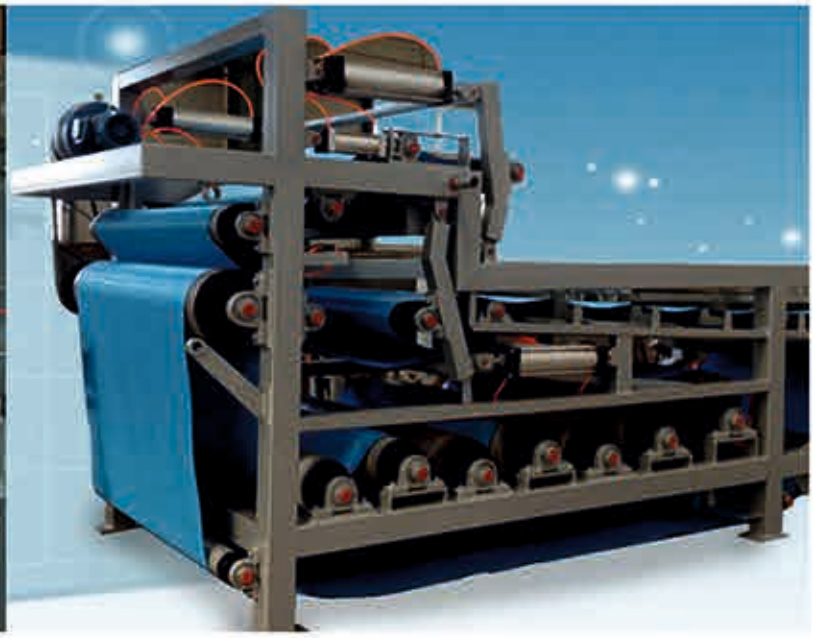
Hydrocyclone

A hydrocyclone is a device to separate or sort particles in a liquid suspension based on the ratio of their centripetal force to fluid resistance. Hydrocyclone will normally have a cylindrical section at the top where liquid is being fed tangentially, and a conical base.









Zero Liquid Discharge

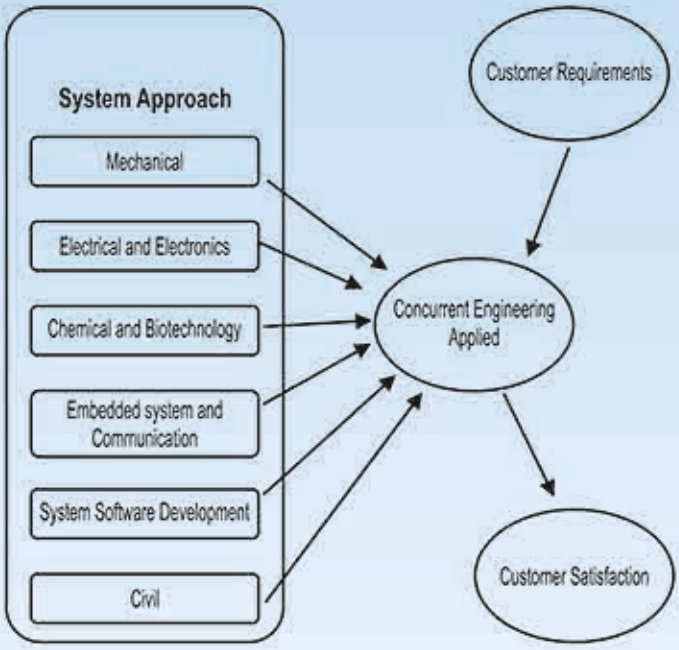
We provide Zero-liquid discharge (ZLD) water treatment scheme with complete automation in which all waste water generated from a running water treatment plant is purified and recycled; therefore, leaving zero discharge at the end of the treatment cycle.

ZLD is an advanced waste water treatment method that includes almost all the operations like aeration, dosing, mixing, sedimentation, solid liquid separation, filtration and disinfection.

Creintors Proposed Zero Liquid Discharge Scheme for River Water Treatment







We, the CREINTORS, have always been the preferred suppliers for Contractors, Government Water Supply and Sewerage Boards at State and National levels. What motivates our clients to partner with us is our philosophy to keep things "Simple" and their faith in our "Quality" products and projects. As we firmly believe –

"THE BEST DESIGN IS ALWAYS SIMPLE
AND A SIMPLE DESIGN THE MOST OPTIMUM".

Achieving this Simple design is tough and requires a team of self-motivated people who can Create, Invent and Innovate. These ingredients in our team makes us
Creators + Inventors = Creintors.

Our strong engineering resource with system approach and team work, always gives us the motivation to perform and ensure that, "We are committed to provide water treatment solutions under one roof, in order to ensure quality treated water and its distribution with total accountability"

come... let's grow

Our success stories...













Crienviro®

Engineering Eco-Sustainability



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